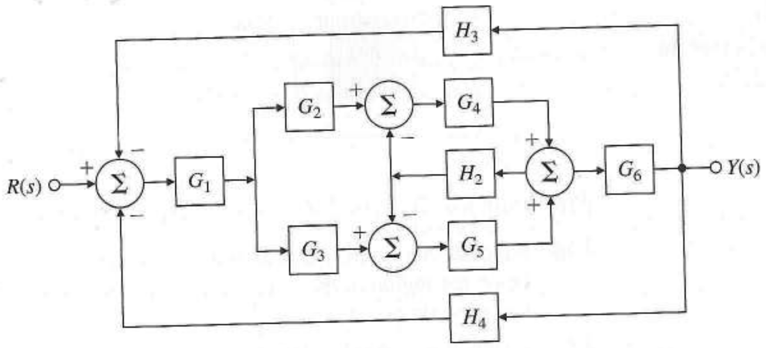
55
gram for
3.22



Problems for Section 3.3: Effect of Pole Locations

3.23 For the electric circuit shown in Fig. 3.56, find the following:

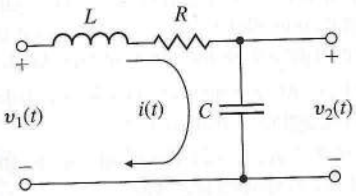
(a) The time-domain equation relating $i(t)$ and $v_1(t)$;

(b) The time-domain equation relating $i(t)$ and $v_2(t)$;

(c) Assuming all initial conditions are zero, the transfer function $\frac{V_2(s)}{V_1(s)}$ and the damping ratio ζ and undamped natural frequency ω_n of the system;

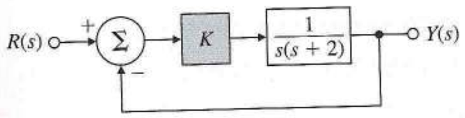
(d) The values of R that will result in $v_2(t)$ having an overshoot of no more than 25%, assuming $v_1(t)$ is a unit step, $L = 10$ mH, and $C = 4\ \mu$F.

3.56
r Problem 3.23



3.24 For the unity feedback system shown in Fig. 3.57, specify the gain K of the proportional controller so that the output $y(t)$ has an overshoot of no more than 10% in response to a unit step.

3.57
feedback system
Problem 3.24



3.25 For the unity feedback system shown in Fig. 3.58, specify the gain and pole location of the compensator so that the overall closed-loop response to a unit-step input has an overshoot of no more than 25%, and a 1% settling time of no more than 0.1 sec. Verify your design using MATLAB.

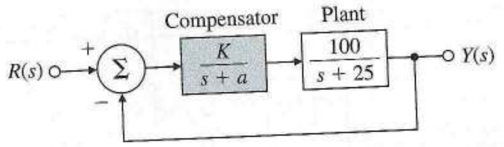


Figure 3.58
Unity feedback system for Problem 3.25

Problems for Section 3.4: Time-Domain Specification

3.26 Suppose you desire the peak time of a given second-order system to be less than t_p'. Draw the region in the s-plane that corresponds to values of the poles that meet the specification $t_p < t_p'$.

3.27 A certain servomechanism system has dynamics dominated by a pair of complex poles and no finite zeros. The time-domain specifications on the rise time (t_r), percent overshoot (M_p), and settling time (t_s) are given by

$$t_r \le 0.6 \text{ sec},$$
$$M_p \le 17\%,$$
$$t_s \le 9.2 \text{ sec}.$$

(a) Sketch the region in the s-plane where the poles could be placed so that the system will meet *all* three specifications.

(b) Indicate on your sketch the specific locations (denoted by $\times$) that will have the smallest rise-time and also meet the settling time specification *exactly*.

3.28 Suppose you are to design a unity feedback controller for a first-order plant depicted in Fig. 3.59. (As you will learn in Chapter 4, the configuration shown is referred to as a proportional-integral controller.) You are to design the controller so that the closed-loop poles lie within the shaded regions shown in Fig. 3.60.

(a) What values of ω_n and ζ correspond to the shaded regions in Fig. 3.59? (A simple estimate from the figure is sufficient.)

(b) Let $K_\alpha = \alpha = 2$. Find values for K and K_I so that the poles of the closed-loop system lie within the shaded regions.

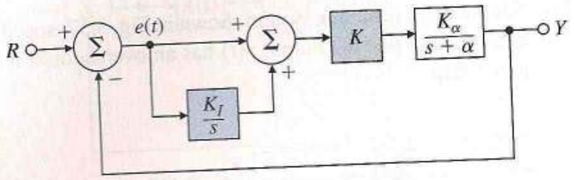


Figure 3.59
Unity feedback system for Problem 3.28

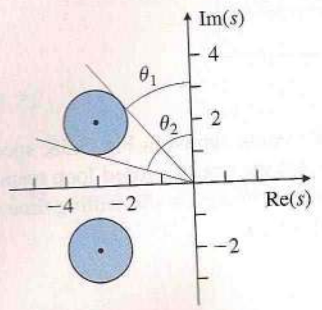


Figure 3.60
Desired closed-loop pole locations for Problem 3.28

(c) Prove that no matter what the values of K_α and α are, the controller provides enough flexibility to place the poles anywhere in the complex (left-half) plane.

3.29 The open-loop transfer function of a unity feedback system is

$$G(s) = \frac{K}{s(s+2)}.$$

The desired system response to a step input is specified as peak time $t_p = 1$ sec and overshoot $M_p = 5\%$.

(a) Determine whether both specifications can be met simultaneously by selecting the right value of K.

(b) Sketch the associated region in the s-plane where both specifications are met, and indicate what root locations are possible for some likely values of K.

△ **(c)** Relax the specifications in part (a) by the same factor and pick a suitable value for K, and use MATLAB to verify that the new specifications are satisfied.

3.30 The equations of motion for the DC motor shown in Fig. 2.32 were given in Eqs. (2.52–2.53) as

$$J_m\ddot{\theta}_m + \left(b + \frac{K_t K_e}{R_a}\right)\dot{\theta}_m = \frac{K_t}{R_a}v_a.$$

Assume that

$$\begin{aligned} J_m &= 0.01\ \text{kg·m}^2, \\ b &= 0.001\ \text{N·m·sec}, \\ K_e &= 0.02\ \text{V·sec}, \\ K_t &= 0.02\ \text{N·m/A}, \\ R_a &= 10\ \Omega. \end{aligned}$$

(a) Find the transfer function between the applied voltage v_a and the motor speed $\dot{\theta}_m$.

(b) What is the steady-state speed of the motor after a voltage $v_a = 10$ V has been applied?

(c) Find the transfer function between the applied voltage v_a and the shaft angle θ_m.

(d) Suppose feedback is added to the system in part (c) so that it becomes a position servo device such that the applied voltage is given by

$$v_a = K(\theta_r - \theta_m),$$

where K is the feedback gain. Find the transfer function between θ_r and θ_m.

(e) What is the maximum value of K that can be used if an overshoot $M_p < 20\%$ is desired?

(f) What values of K will provide a rise time of less than 4 sec? (Ignore the M_p constraint.)

(g) Use MATLAB to plot the step response of the position servo system for values of the gain $K = 0.5$, 1, and 2. Find the overshoot and rise time for each of the three step responses by examining your plots. Are the plots consistent with your calculations in parts (e) and (f)?

3.31 You wish to control the elevation of the satellite-tracking antenna shown in Figs. 3.61 and 3.62. The antenna and drive parts have a moment of inertia J and a damping B;